A Bayesian Statistical Framework for Detecting Engine-Assisted Play in Online Gomoku Competitions

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Abstract

The integrity of online board game competitions is increasingly threatened by players who covertly use artificial intelligence engines during live matches. This paper presents a rigorous Bayesian statistical framework for detecting engine-assisted play in Gomoku (Five-in-a-Row), a combinatorial strategy game. Our approach models the *winrate delta*—the difference between the engine-optimal move evaluation and the player's actual move evaluation—as a random variable drawn from parametric distributions that differ between legitimate human players and engine-assisted players. We develop the framework within a general family of non-negative distributions (Exponential, Gamma, Weibull), and introduce a *two-component mixture model* to capture the bimodal nature of human play, with a Dirichlet Process extension for automatic component selection. A *context-aware complexity metric* weights evidence by positional difficulty, and a *tempered likelihood* mechanism prevents trivial positions from distorting posterior estimates. The framework performs sequential Bayesian updates over the course of a game, integrating a Boltzmann choice model, information-theoretic divergence measures, and temporal switch-point detection to identify mid-game engine activation. We further introduce *player skill stratification* to address the false-positive problem arising from strong legitimate players, a formal *multi-game profile analysis* grounded in confidence-interval theory and hypothesis testing for population-level inference, and an *opponent-adjusted complexity* mechanism that corrects for positions made artificially easy by a weak opponent—a previously unaddressed source of false-positive inflation. A formal likelihood ratio test grounded in Neyman–Pearson theory provides false-positive rate control. We describe the complete mathematical formulation, the system architecture for engine integration, and the classification decision boundaries, and report preliminary experimental observations that motivate several design choices.

Keywords: Anti-cheating, Bayesian inference, Gomoku, engine-assisted play detection, online gaming integrity, mixture model, Gamma distribution, Weibull distribution, Dirichlet process, information theory, temporal analysis, skill stratification, profile analysis, opponent-adjusted complexity.

1 Introduction

Online board game platforms have experienced significant growth, enabling players worldwide to compete in real-time tournaments. However, the remote and unsupervised nature of online play creates an avenue for a specific form of cheating: covertly consulting an AI engine during a live match. In Gomoku (also known as Five-in-a-Row or Renju in its professional variant), modern engines achieve superhuman playing strength, making engine-assisted play both highly effective and difficult to distinguish from expert human play through casual observation alone.

Existing anti-cheating measures in online gaming have largely relied on behavioral heuristics—such as timing patterns, mouse movement tracking, or tab-switching detection—which are easily

circumvented by sophisticated cheaters. Statistical approaches have been explored in chess [Regan and Haworth, 2011], but relatively little formal work has been published for Gomoku and its variants. Furthermore, many deployed systems rely on ad-hoc thresholds (e.g., "if accuracy exceeds 90%, flag the player") without a coherent probabilistic foundation, leading to both false positives against strong legitimate players and false negatives against selective cheaters.

A fundamental challenge, which we term the *skill-confounding problem*, arises because a single-population null hypothesis H_0 cannot simultaneously represent beginners, intermediate players, and professionals. A professional player's move quality may exceed the system's cheater threshold while being entirely consistent with legitimate play. Preliminary experiments (Section 11) confirm this: a professional player analyzed over 22 moves yielded $\hat{\lambda}_{\text{MLE}} = 113$, far exceeding the default cheater threshold of $\lambda_{\text{cheater}} = 50$, yet the player was confirmed to be playing without assistance. This motivates the stratified null hypothesis developed in Section 3.3.

This paper addresses these limitations by developing a complete Bayesian statistical framework that:

(i) Models human and cheater move-quality distributions using parametric families grounded in statistical theory.

(ii) Incorporates *positional context* through a complexity metric that modulates the evidential weight of each move.

(iii) Performs online (sequential) Bayesian updates with tempered likelihoods, ensuring that trivial positions do not corrupt the posterior estimate.

(iv) Employs information-theoretic divergence measures and temporal change-point detection to identify partial or mid-game engine use.

(v) Stratifies the null hypothesis by player skill level and aggregates evidence across multiple games via a formal profile analysis.

(vi) Corrects for *opponent-induced easy positions* via an opponent-adjusted complexity mechanism that prevents artificially inflated posteriors when the analyzed player faces a weak opponent.

1.1 Contributions

The main contributions of this work are:

1. **Winrate Delta Model.** We formalize the concept of *winrate delta* and establish that the Exponential distribution family provides a natural parametric model for move quality under both hypotheses, with Gamma and Weibull as empirically motivated generalizations.

2. **Two-Component Mixture Model.** We propose a mixture of Exponentials for the human hypothesis that accounts for the bimodal nature of human play (careful moves interspersed with occasional blunders), with parameters estimated via the EM algorithm.

3. **Player Skill Stratification.** We introduce a stratified null hypothesis that conditions on the player's estimated skill tier, resolving the false-positive problem for strong legitimate players.

4. **Context-Aware Complexity.** We derive a composite complexity metric from candidate move evaluations that weights Bayesian evidence by positional difficulty, preventing trivial positions from generating misleading signals.

5. **Tempered Bayesian Updates.** We introduce a tempering mechanism where the likelihood exponent equals the positional complexity, providing a mathematically principled method for evidence weighting.

6. **Multi-Game Profile Analysis.** We develop a formal population-level inference framework that aggregates evidence across multiple games, with sample size requirements derived from confidence-interval theory.

7. **Temporal Switch-Point Detection.** We develop a temporal model that detects abrupt changes in play quality, identifying possible mid-game engine activation.

8. **Opponent-Adjusted Complexity.** We identify and formalize the *opponent-difficulty confound*—the systematic inflation of false-positive rates when the analyzed player faces a weak opponent— and introduce the Opponent Difficulty Factor to correct for it.

9. **Complete System Design.** We describe the full architecture for integrating an external game engine as an evaluation oracle within the detection pipeline.

1.2 Paper Organization

Section 2 reviews related work. Section 3 presents the mathematical framework in detail. Section 4 develops the context-aware complexity model. Section 5 describes the Bayesian inference engine. Section 6 introduces the information-theoretic metrics. Section 7 covers temporal analysis and switch-point detection. Section 8 develops the multi-game profile analysis framework. Section 9 outlines the system architecture. Section 10 defines the classification decision boundaries. Section 11 reports preliminary experimental observations. Section 12 discusses strengths, limitations, and ethical considerations. Section 13 concludes the paper.

2 Related Work

2.1 Engine-Assisted Play Detection in Chess

The most extensively studied domain for engine-assisted play detection is chess. Regan and Haworth [Regan and Haworth, 2011] proposed an intrinsic performance rating system that estimates a player's Elo rating from their moves alone, comparing observed play quality to engine evaluations. Their approach uses a logistic model to predict the probability that a player of a given rating would select each candidate move. Significant deviations between the predicted rating and the player's registered rating are flagged as suspicious.

The *Let's Check* system used by ChessBase and similar platforms [Guid and Bratko, 2006] compares player moves against deep engine analysis, computing aggregate statistics such as average centipawn loss and correlation with engine top choices. While effective as a screening tool, these systems typically rely on fixed thresholds rather than probabilistic inference.

2.2 Statistical Methods for Fair Play

Broader statistical approaches to fair play detection include hypothesis testing frameworks [Berger and Sellke, 1987], where the null hypothesis is honest play and the alternative is engine assistance. The challenge lies in controlling the false positive rate—strong human players naturally exhibit high move quality, and a framework that flags top-level play as suspicious undermines competitive integrity.

2.3 Gomoku-Specific Considerations

Gomoku differs from chess in several important respects that affect anti-cheating analysis. The game has a narrower decision tree in many positions due to forced tactical sequences (threats, double-threats), meaning that many "perfect" moves are also the obvious human choice. A robust detection framework must therefore distinguish between positions where engine-optimal play is trivially findable and positions where it requires deep calculation. This motivates our development of the context-aware complexity model (Section 4).

3 Mathematical Framework: The Winrate Delta Model

3.1 Definitions

We assume access to an external game engine that, given a board position s, produces a set of k candidate moves $\{m_1, m_2, \ldots, m_k\}$ along with their associated *winrate evaluations* $\{w_1, w_2, \ldots, w_k\}$, where $w_i \in [0, 1]$ represents the probability of winning from the resulting position if move m_i is played.

Definition 1 (Winrate Delta)**.** *For a given position, let $w^* = \max_i w_i$ denote the winrate of the engine's best candidate move, and let w_p denote the winrate of the move actually played by the player. The* winrate delta *is defined as:*

$$\Delta = w^* - w_p, \quad \Delta \geq 0. \tag{1}$$

The winrate delta Δ captures the *quality loss* incurred by the player's move choice. A Δ of zero indicates the player found the engine-optimal move; a large Δ indicates a suboptimal choice.

Remark 1. *All winrates are expressed from the perspective of the player being analyzed. If the engine reports winrates from the opponent's perspective, the inversion $w_{player} = 1 - w_{opponent}$ must be applied before computing Δ.*

3.2 Distributional Family for the Winrate Delta

Because $\Delta \geq 0$ by definition, the appropriate parametric models belong to the family of non-negative continuous distributions. We consider three candidates in order of increasing flexibility.

3.2.1 Exponential Distribution (Baseline)

The simplest model assumes $\Delta \sim \text{Exponential}(\lambda)$:

$$f(\Delta \mid \lambda) = \lambda\, e^{-\lambda \Delta}, \quad \Delta \geq 0. \tag{2}$$

Its memoryless property implies that the magnitude of a suboptimal choice does not depend on previous errors conditional on the current position. However, this assumption may be violated when players exhibit psychological momentum ("tilt" or "flow state"), producing serial correlation in error magnitudes.

3.2.2 Gamma Distribution (Shape-Rate Generalization)

$$f(\Delta \mid \alpha, \beta) = \frac{\beta^\alpha\, \Delta^{\alpha-1}\, e^{-\beta\Delta}}{\Gamma(\alpha)}, \quad \Delta \geq 0. \tag{3}$$

When $\alpha = 1$, the Gamma reduces to Exponential(β). For $\alpha > 1$, the density is unimodal with mode at $(\alpha - 1)/\beta$, enabling modeling of delta distributions that concentrate around a typical error magnitude. The mean and variance are $\mathbb{E}[\Delta] = \alpha/\beta$ and $\text{Var}(\Delta) = \alpha/\beta^2$.

3.2.3 Weibull Distribution (Tail Flexibility)

$$f(\Delta \mid k, \lambda) = \frac{k}{\lambda}\left(\frac{\Delta}{\lambda}\right)^{k-1} \exp\left[-\left(\frac{\Delta}{\lambda}\right)^{k}\right], \quad \Delta \geq 0. \tag{4}$$

The shape parameter k controls tail weight: $k < 1$ yields a heavier tail than the Exponential (suitable for humans prone to rare severe blunders), while $k > 1$ yields a lighter tail (suitable for cheater modeling). Preliminary experiments consistently select Weibull as the best-fit model by AIC/BIC (Section 11), though all models are rejected by goodness-of-fit tests at small sample sizes ($n \leq 30$), underscoring the need for multi-game aggregation.

3.2.4 Model Selection via Information Criteria

$$\mathrm{AIC} = 2k - 2\ln \hat{L}, \tag{5}$$

$$\mathrm{BIC} = k \ln n - 2\ln \hat{L}. \tag{6}$$

Lower values indicate better model–data fit after penalizing complexity. Goodness-of-fit tests (Kolmogorov–Smirnov or Anderson–Darling) should additionally be applied to validate distributional assumptions before deployment.

3.3 Hypothesis Structure and Player Skill Stratification

A critical deficiency of single-population null hypothesis models is the *skill-confounding problem*: a professional player's move quality may be statistically indistinguishable from engine-assisted play under a model calibrated for average-strength players. We address this by introducing a *stratified null hypothesis.*

3.3.1 Skill Tiers

We partition the player population into skill tiers $\mathcal{T} = \{T_1, T_2, T_3, T_4\}$, each characterized by a distinct null distribution parameter:

Table 1: Proposed skill tier parameterization for the Exponential baseline. Values should be calibrated empirically from labeled reference data.

Tier	Description	λ_{H_0} (indicative)	$\mathbb{E}[\Delta]$
T_1	Beginner	3–8	12–33%
T_2	Intermediate	10–25	4–10%
T_3	Advanced	30–60	1.7–3.3%
T_4	Professional	80–150	0.7–1.3%
H_1	Engine-assisted	200–500	0.2–0.5%

Remark 2. *The values in Table 1 are indicative only. Empirical calibration from large labeled datasets (Section 3.6) is required before deployment. A key observation from preliminary experiments is that a professional player's $\hat{\lambda}_{MLE}$ consistently falls in the range* 90–120 *(see Section 11), firmly within tier T_4, while the current default $\lambda_{cheater} = 50$ lies* within *the professional human range—demonstrating the necessity of stratification.*

3.3.2 Tier Assignment

In practice, the player's tier is assigned based on one of three methods, in decreasing order of reliability:

1. **Profile-estimated tier:** The player's historical $\hat{\lambda}$ from confirmed honest games (Section 8) directly determines tier membership.

2. **Registered rating:** The player's tournament rating (Elo or equivalent) maps to a tier via calibrated thresholds.

3. **Prior tier:** In the absence of historical data, a weakly informative prior assigns the player to the most common tier for the relevant competitive context.

3.3.3 Stratified Likelihood

Under the stratified null hypothesis, the human likelihood in Equation (29) is replaced by the tier-conditional likelihood:

$$P(\Delta \mid H_0) = P(\Delta \mid H_0, T^*), \tag{7}$$

where T^* is the player's assigned tier. This ensures that the Bayesian posterior compares the observed play to the player's *own peer group* rather than the overall player population.

3.4 Maximum Likelihood Estimation

Given a sequence of observed deltas $\Delta_1, \Delta_2, \ldots, \Delta_n$, the MLE for the Exponential rate parameter is:

$$\hat{\lambda}_{\text{MLE}} = \frac{n}{\sum_{i=1}^{n} \Delta_i} = \frac{1}{\bar{\Delta}}. \tag{8}$$

To handle $\Delta_i = 0$, we apply a floor $\epsilon = 0.001$:

$$\Delta_i^{\text{adj}} = \max(\Delta_i,\, \epsilon). \tag{9}$$

We define the *Accuracy Index* as:

$$\text{AccuracyIndex} = \frac{1}{\hat{\lambda}_{\text{MLE}}} = \bar{\Delta}^{\text{adj}}, \tag{10}$$

which directly measures the average winrate loss. Lower values indicate play closer to engine-optimal.

3.5 Two-Component Exponential Mixture for Human Play

A single Exponential distribution inadequately captures human play, which exhibits a bimodal pattern: the majority of moves are sound (small Δ), but occasional moves are blunders (large Δ). We model the human delta distribution as:

$$P(\Delta \mid H_0) = \pi \cdot \lambda_g\, e^{-\lambda_g \Delta} + (1-\pi) \cdot \lambda_b\, e^{-\lambda_b \Delta}, \tag{11}$$

where $\pi \in (0, 1)$ is the mixing weight for the "good play" component, λ_g is the rate for good plays, and λ_b is the rate for blunders.

The log-sum-exp trick ensures numerical stability:

$$a = \log \pi + \log \lambda_g - \lambda_g \Delta, \tag{12}$$

$$b = \log(1-\pi) + \log \lambda_b - \lambda_b \Delta, \tag{13}$$

$$\log P(\Delta \mid H_0) = m + \log\left(e^{a-m} + e^{b-m}\right), \tag{14}$$

where $m = \max(a, b)$.

3.6 Parameter Estimation via Expectation-Maximization

The mixture parameters $\Theta = (\pi, \lambda_g, \lambda_b)$ are estimated from labeled reference data via the EM algorithm [Dempster et al., 1977].

E-step:

$$\gamma_{i,1} = \frac{\pi \cdot \lambda_g e^{-\lambda_g \Delta_i}}{\pi \cdot \lambda_g e^{-\lambda_g \Delta_i} + (1-\pi) \cdot \lambda_b e^{-\lambda_b \Delta_i}}, \quad \gamma_{i,2} = 1 - \gamma_{i,1}. \tag{15}$$

M-step:

$$\pi^{\text{new}} = \frac{1}{n} \sum_{i=1}^{n} \gamma_{i,1}, \tag{16}$$

$$\lambda_g^{\text{new}} = \frac{\sum_i \gamma_{i,1}}{\sum_i \gamma_{i,1} \Delta_i}, \quad \lambda_b^{\text{new}} = \frac{\sum_i \gamma_{i,2}}{\sum_i \gamma_{i,2} \Delta_i}. \tag{17}$$

Remark 3. *When EM is applied to a single-game dataset ($n \leq 30$), the fitted parameters exhibit high variance and should be treated as exploratory descriptors rather than calibrated estimates. Reliable EM convergence requires $n \geq 200$ observations (Section 8.4).*

3.7 Dirichlet Process Mixture Model

The fixed two-component mixture assumes exactly two behavioral modes. The *Dirichlet Process Mixture Model* (DPMM) [Ferguson, 1973, Neal, 2000] relaxes this assumption:

$$G \sim \text{DP}(\alpha_0, G_0), \qquad \theta_i \mid G \sim G, \qquad \Delta_i \mid \theta_i \sim F(\cdot \mid \theta_i). \tag{18}$$

The effective number of components is governed by the concentration parameter α_0 and inferred from data, avoiding model selection entirely.

3.8 Boltzmann Choice Model

We model the player's move selection using a Boltzmann distribution:

$$P(m_i \mid \{w_j\}, \tau) = \frac{\exp(w_i/\tau)}{\sum_{j=1}^{k} \exp(w_j/\tau)}, \tag{19}$$

where $\tau \to 0$ concentrates mass on the highest-winrate move (engine-like) and $\tau \to \infty$ produces a uniform distribution (random play).

The temperature is estimated by maximum likelihood:

$$\hat{\tau} = \arg\max_{\tau} \sum_{i=1}^{n} \log P(m_{c_i} \mid \{w_j^{(i)}\}, \tau). \tag{20}$$

The temperature score $S_\tau = 1/\hat{\tau}$ enters the classification pipeline as an auxiliary signal (Section 10).

4 Context-Aware Position Complexity

4.1 Delta Vector and Key Metrics

$$d_i = w_i - w_{\text{prev}}, \quad d^* = \max_i d_i, \quad \bar{d} = \frac{1}{k} \sum_i d_i, \quad S = d^* - \min_i d_i. \tag{21}$$

4.2 Complexity Factors

Definition 2 (Impact Factor)**.** $F_{impact} = \min(|d^*|/0.25,\ 1.0)$.

Definition 3 (Variance Factor)**.** $F_{variance} = \min(S/0.40,\ 1.0)$.

Definition 4 (Criticality Factor)**.** $F_{criticality} = (d^* - \bar{d})/S$ *if* $S > 0.01$*, else* 0*, clamped to* $[0, 1]$.

Definition 5 (Phase Factor)**.**

$$F_{phase} = \begin{cases} 0.3 & \text{if } w^* > 0.85, \\ 0.2 & \text{if } w^* < 0.15, \\ 0.5 + 0.5(1 - 2|w^* - 0.5|) & \text{otherwise.} \end{cases} \tag{22}$$

4.3 Combined Complexity Score

$$C = 0.15F_{\text{impact}} + 0.40F_{\text{variance}} + 0.20F_{\text{criticality}} + 0.25F_{\text{phase}}. \tag{23}$$

Proposition 1 (Trivial Position Override)**.** *If* $w^* > 0.95$ *or* $w^* < 0.05$ *and* $F_{variance} < 0.05$*, then* $C = 0.10$.

4.4 Opponent-Adjusted Complexity

The complexity score C defined above depends solely on the *analyzed player's* candidate move evaluations, and is therefore blind to the state of the opponent's position. This creates a systematic bias that we term the *opponent-difficulty confound*.

Definition 6 (Opponent-Difficulty Confound)**.** *When the analyzed player's opponent is in a severely losing position, many of the player's candidate moves achieve near-optimal winrate (because the opponent has no effective defense). A one-sided complexity model that ignores opponent position assigns non-negligible complexity* C *to these positions, incorrectly treating trivially-optimal play as informative evidence of engine assistance.*

Remark 4 (Empirical Motivation)**.** *In a preliminary game analysis (Section 11.3), multiple late-game positions were assigned complexity* $C \approx 0.49$ *despite the opponent's best available move having a winrate below* 0.3%*. The Trivial Position Override (Proposition 1) failed to activate because* w^* *had not yet exceeded the* 0.95 *threshold from the player's perspective, while the opponent was already effectively defeated. These positions contributed spurious evidence toward* H_1.

4.4.1 Opponent Difficulty Factor

Let w^*_{opp} denote the best available winrate for the opponent at position t, expressed from the *opponent's perspective* (i.e., already normalized via $w_{\text{opp}} = 1 - w_{\text{player}}$ per Remark 1). The *Opponent Difficulty Factor* is defined as:

$$F_{\text{opp}} = \min\left(\frac{w^*_{\text{opp}}}{w_{\text{ref}}},\ 1.0\right), \tag{24}$$

where $w_{\text{ref}} = 0.40$ is a calibration threshold representing the winrate below which the opponent is no longer considered a meaningful competitive threat. The factor $F_{\text{opp}} \in [0, 1]$ takes value 1 when the opponent is fully competitive and decreases toward 0 as the opponent's position becomes hopeless.

Remark 5 (Temporal Indexing)**.** *To avoid circular dependency,* w^*_{opp} *at position* t *is evaluated* after the opponent's most recent move *and* before *the analyzed player's current move is played. This ensures that we measure the opponent's strength at the moment the decision is made, not the consequence of the player's own move.*

4.4.2 Opponent-Adjusted Complexity Score

The raw complexity C (Equation (23)) is adjusted as:

$$C_{\text{adj}} = C \cdot \max(F_{\text{opp}},\ \alpha_{\min}), \tag{25}$$

where $\alpha_{\min} = 0.10$ ensures that a minimum residual weight is retained even in extreme cases, preserving the continuity of the evidence accumulation process.

The final complexity used in all downstream computations (tempered likelihood, context-aware accuracy) is:

$$C_{\text{final}} = \begin{cases} 0.10 & \text{if Trivial Position Override is triggered,} \\ C_{\text{adj}} & \text{otherwise.} \end{cases} \tag{26}$$

Remark 6 (Effect on Tempered Likelihood)**.** *Note that C_{final} acts as the tempering exponent α in Equation (33). Therefore, when the opponent is nearly defeated ($F_{opp} \approx 0$), the tempered likelihood approaches $P(\Delta \mid H)^{\alpha_{\min}} = P(\Delta \mid H)^{0.10}$, contributing only minimal evidence to the Bayesian posterior regardless of how small Δ is. This is the mathematically correct behavior: a perfect move in a trivially-won position reveals nothing about the player's computational capability.*

Table 2 illustrates the effect of the adjustment on representative positions from the experimental game log.

Table 2: Effect of Opponent-Adjusted Complexity on selected moves (experimental game, Black side). C denotes raw complexity; F_{opp} denotes the Opponent Difficulty Factor; C_{adj} denotes the adjusted complexity.

Move	Δ	C	w^*_{opp}	F_{opp}	C_{adj}	Opp Quality
#7	0.0%	0.56	36.9%	0.92	0.515	Passive (competitive)
#9	3.5%	0.46	36.4%	0.91	0.419	Tactical (competitive)
#11	0.0%	0.62	22.9%	0.57	0.355	Tactical (weakening)
#13	0.0%	0.47	0.2%	**0.005**	**0.047**	Winning (trivial)
#33	0.0%	0.49	0.2%	**0.005**	**0.049**	Winning (trivial)
#35	0.0%	0.10	0.2%	0.005	0.010	Override applied

Without the adjustment, moves #13 and #33 contribute $C = 0.47$–0.49 worth of evidence toward H_1 despite occurring in trivially-decided positions. With C_{adj}, their evidential contribution drops to ≈ 0.05, commensurate with their actual informativeness.

4.5 Context-Aware Accuracy

All accuracy and likelihood computations use C_{final} (Equation (26)) in place of the raw complexity C:

$$A(\Delta, C_{\text{final}}) = \begin{cases} 1.0 & \text{if } \Delta \leq 0.02, \\ \exp(-5\Delta(1 - 0.3C_{\text{final}})) & \text{otherwise.} \end{cases} \tag{27}$$

4.5.1 Strategic Bonus

$$A_{\text{final}} = A \cdot (1 + 0.15C_{\text{opp}}), \quad \text{if } A > 0.4 \text{ and } C_{\text{opp}} > 0.1, \tag{28}$$

clamped to $[0, 1]$.

5 Bayesian Inference Engine

5.1 Posterior Computation

$$P(H_1 \mid D) = \frac{P(D \mid H_1) \cdot P(H_1)}{P(D \mid H_1) \cdot P(H_1) + P(D \mid H_0) \cdot P(H_0)}, \tag{29}$$

with $P(H_1) = 0.01$ (conservative prior) and $P(H_0) = 1 - P(H_1)$.

In log space:

$$L_0 = \sum_{i=1}^{n} \log P(\Delta_i \mid H_0), \qquad L_1 = \sum_{i=1}^{n} \log P(\Delta_i \mid H_1), \tag{30}$$

$$P(H_1 \mid D) = \frac{e^{\ell_1 - m}}{e^{\ell_0 - m} + e^{\ell_1 - m}}, \tag{31}$$

where $\ell_j = L_j + \log P(H_j)$ and $m = \max(\ell_0, \ell_1)$.

5.2 Online (Sequential) Bayesian Updates

$$P(H_1 \mid D_{1:n+1}) \propto P(\Delta_{n+1} \mid H_1)^{C_{n+1}} \cdot P(H_1 \mid D_{1:n}). \tag{32}$$

5.3 Tempered Likelihood

We raise the likelihood to the power of the positional complexity C:

$$P(\Delta \mid H)^C \quad \Longleftrightarrow \quad C \cdot \log P(\Delta \mid H). \tag{33}$$

When $C = 0$ (trivial position), the tempered likelihood is 1, contributing zero evidence. When $C = 1$, full evidence is contributed. This mechanism is well-studied in the statistical literature [Grünwald and van Ommen, 2017].

Algorithm 1 Online Bayesian Update with Tempered Likelihood

Require: Current posterior p, new delta Δ, complexity C
Ensure: Updated posterior p'

```
if C ≤ 0 then
    return p
end if
ℓ0 ← C · log P(Δ | H0) + log(1 − p)
ℓ1 ← C · log P(Δ | H1) + log(p)
m ← max(ℓ0, ℓ1)
return e^(ℓ1−m) / (e^(ℓ0−m) + e^(ℓ1−m))
```

5.4 Confidence Estimation

$$\text{confidence} = \min\left(1, \frac{n}{n_0}\right) \times 2\,|P(H_1 \mid D) - 0.5|\,, \tag{34}$$

where n_0 is a data-sufficiency threshold. The value $n_0 = 30$ used in the original formulation is a lower bound; Section 8.4 derives principled values based on the desired estimation precision. We recommend $n_0 = 44$ for proportion-based classification and $n_0 = 97$ for λ-based inference.

5.5 Formal Likelihood Ratio Test

$$\Lambda = \frac{P(D \mid H_1)}{P(D \mid H_0)} = \exp(L_1 - L_0). \tag{35}$$

By the Neyman–Pearson lemma [Neyman and Pearson, 1933], the test that rejects H_0 when $\Lambda > \eta$ is the most powerful test at significance level α. The critical value η is calibrated from a reference corpus of confirmed human games.

5.6 Prior Sensitivity Analysis

We evaluate $P(H_1 \mid D)$ across $P(H_1) \in \{0.001, 0.01, 0.05, 0.10\}$. If the posterior exceeds the cheater threshold for all reasonable priors, the evidence is considered robust. Sensitivity analysis is standard practice in Bayesian applications with subjective priors [Berger and Sellke, 1987].

6 Information-Theoretic Metrics

6.1 Shannon Entropy of Move Quality

$$H(P) = -\sum_i p_i \log_2 p_i, \tag{36}$$

where p_i is the empirical probability of the player's delta falling in bin i. Human players exhibit higher entropy (more varied quality); engine-assisted players exhibit lower entropy (consistently near-optimal).

6.2 KL Divergence from Engine Policy

$$D_{\mathrm{KL}}(P_{\text{player}} \| P_{\text{engine}}) = \sum_i P_{\text{player}}(m_i) \log_2 \frac{P_{\text{player}}(m_i)}{P_{\text{engine}}(m_i)}. \tag{37}$$

Remark 7 (Information Overlap with the Delta Model)**.** *Since P_{player} is one-hot, the KL divergence simplifies to $D_{KL} = -\log_2 Q(m_{chosen})$, which is the negative log-likelihood of the chosen move under the engine's policy. This is informationally related to Δ: both measure engine alignment in different spaces (probability space vs. winrate space). These metrics are* not independent *signals, and practitioners should account for this correlation to avoid overconfident conclusions.*

6.3 Cross-Entropy

$$H(P, Q) = -\sum_i P(m_i) \log_2 Q(m_i). \tag{38}$$

Since $D_{\mathrm{KL}}(P \| Q) = H(P, Q) - H(P)$, cross-entropy provides an unnormalized measure of engine alignment.

7 Temporal Analysis and Switch-Point Detection

7.1 Moving Average Smoothing

$$\bar{\Delta}_t = \frac{1}{\min(t, W)} \sum_{j=\max(1,\, t-W+1)}^{t} \Delta_j. \tag{39}$$

7.2 Switch-Point Detection

A switch point is detected at index t if:

$$|\mu_{\text{before}} - \mu_{\text{after}}| \geq \theta_{\text{switch}}, \tag{40}$$

where μ_{before} and μ_{after} are local means of the smoothed sequence. A positive change (deltas decreased) indicates suspicious improvement.

7.3 Trend and Volatility

Trend is estimated via linear regression slope $\hat{\beta}$ (Equation (41)); volatility via the coefficient of variation $V = \sigma_\Delta / \bar{\Delta}$.

$$\hat{\beta} = \frac{\sum_{i=1}^{n}(i - \bar{i})(\Delta_i - \bar{\Delta})}{\sum_{i=1}^{n}(i - \bar{i})^2}. \tag{41}$$

8 Multi-Game Profile Analysis

A single game provides insufficient statistical power for reliable detection, as demonstrated in Section 11. This section develops a formal framework for aggregating evidence across m games in a player's profile history.

8.1 Profile Dataset Structure

Let $\mathcal{G} = \{g_1, g_2, \ldots, g_m\}$ denote a player's profile history of m games, where game g_j yields estimated rate $\hat{\lambda}_j$ and observed mean delta $\bar{\Delta}_j$. The profile dataset treats each game as a sample unit, yielding the profile statistics:

$$\bar{\lambda} = \frac{1}{m}\sum_{j=1}^{m}\hat{\lambda}_j, \tag{42}$$

$$s_\lambda = \sqrt{\frac{1}{m-1}\sum_{j=1}^{m}(\hat{\lambda}_j - \bar{\lambda})^2}. \tag{43}$$

The profile mean $\bar{\lambda}$ serves as an empirical estimate of the player's true skill level, while s_λ quantifies inter-game variability.

8.2 Confidence Interval for the Baseline Parameter

Under the Central Limit Theorem (applicable for $m \geq 30$, or t-distribution for $m < 30$), the $100(1-\alpha)\%$ confidence interval for the player's true rate λ_{true} is:

$$\bar{\lambda} - t_{\alpha/2,\, m-1} \cdot \frac{s_\lambda}{\sqrt{m}} \;\leq\; \lambda_{\text{true}} \;\leq\; \bar{\lambda} + t_{\alpha/2,\, m-1} \cdot \frac{s_\lambda}{\sqrt{m}}, \tag{44}$$

where $t_{\alpha/2,\, m-1}$ is the upper $\alpha/2$ critical value of the t-distribution with $m-1$ degrees of freedom.

8.3 Anomaly Detection via Hypothesis Test

When analyzing a suspect game with rate estimate $\hat{\lambda}_{\text{suspect}}$, we formulate a one-sided test:

$$H_0 : \lambda_{\text{suspect}} = \bar{\lambda}, \tag{45}$$

$$H_1 : \lambda_{\text{suspect}} > \bar{\lambda}. \tag{46}$$

The test statistic is:

$$Z = \frac{\hat{\lambda}_{\text{suspect}} - \bar{\lambda}}{s_\lambda / \sqrt{m}}. \tag{47}$$

We reject H_0 at level α if $Z > z_\alpha$ (or $Z > t_{\alpha,\, m-1}$ for small m). This provides a formally calibrated decision rule whose false-positive rate is controlled at α, as opposed to the ad-hoc posterior thresholds in Equation (51).

8.4 Sample Size Requirements

We derive formal sample size requirements for the profile analysis from confidence-interval theory [Montgomery and Runger, 2010].

8.4.1 Number of Games Required

To estimate λ_{true} with absolute margin E at $100(1-\alpha)\%$ confidence:

$$m \geq \left(\frac{z_{\alpha/2} \cdot s_\lambda}{E}\right)^2. \tag{48}$$

Using the pilot estimate $s_\lambda \approx 12$ obtained from preliminary experiments (Section 11):

Table 3: Required number of games m for estimating λ_{true} at 95% confidence ($z_{0.025} = 1.96$, $s_\lambda = 12$).

Margin E	Formula	Required m
± 10	$\lceil (1.96 \times 12/10)^2 \rceil$	6
± 5	$\lceil (1.96 \times 12/5)^2 \rceil$	22
± 3	$\lceil (1.96 \times 12/3)^2 \rceil$	62

8.4.2 Number of Moves per Game

For estimating the near-optimal proportion p (a binomial parameter) with margin E at 95% confidence, using the pilot estimate $\hat{p} \approx 0.87$:

$$n_{\text{moves}} \geq \left(\frac{z_{\alpha/2}}{E}\right)^2 \hat{p}(1-\hat{p}). \tag{49}$$

Table 4: Required moves per game for estimating near-optimal proportion at 95% confidence ($\hat{p} = 0.87$).

Margin E	Formula	Required n
± 0.10	$\lceil (19.6)^2 \times 0.113 \rceil$	44
± 0.05	$\lceil (39.2)^2 \times 0.113 \rceil$	174
± 0.025	$\lceil (78.4)^2 \times 0.113 \rceil$	694

For λ-based inference, the MLE variance of the Exponential rate parameter satisfies $\text{SE}(\hat{\lambda}) = \lambda/\sqrt{n}$, yielding the relative-error requirement:

$$n \geq \left(\frac{z_{\alpha/2}}{E_{\text{rel}}}\right)^2, \tag{50}$$

where E_{rel} is the desired relative error. For $E_{\text{rel}} = 0.20$ and 95% confidence: $n \geq (1.96/0.20)^2 = 97$ moves.

8.4.3 Practical Recommendations

Table 5: Practical sample size guidelines.

Objective	Minimum	Recommended
Confidence score $\geq 80\%$ (single game)	30 moves	44 moves
Proportion estimate $\pm 10\%$	44 moves	100 moves
λ estimate $\pm 20\%$	97 moves	200 moves
EM calibration (mixture model)	200 moves	500 moves
Profile baseline (± 10 in λ)	6 games	20 games
Profile baseline (± 5 in λ)	22 games	30 games

8.5 Stationarity Assumption and Rolling Window

The profile analysis rests on the assumption that the player's true skill level is approximately *stationary* over the observation window.

Remark 8 (Stationarity Assumption)**.** *The profile mean $\bar{\lambda}$ is a valid estimator of λ_{true} only if the player's skill level does not change systematically over the analysis window. This assumption may be violated in two scenarios: (i) genuine skill improvement, which causes $\hat{\lambda}_j$ to increase over time due to legitimate learning; and (ii) onset of engine use at a specific historical game, which causes a structural break in the series $\hat{\lambda}_1, \ldots, \hat{\lambda}_m$.*

To address this, we recommend a *rolling window* strategy: only the m_{window} most recent games are used to estimate the baseline, with $m_{\text{window}} \in [20, 50]$ as a practical range. The temporal switch-point detection of Section 7 can be applied to the game-level series $\hat{\lambda}_1, \ldots, \hat{\lambda}_m$ to detect structural breaks in player skill, which may themselves constitute suspicious patterns.

9 System Architecture

9.1 Overview

The detection system consists of three layers:

1. **Engine Layer**: An external Gomoku engine (e.g., Rapfi) communicates via the Gomocup protocol. For each position, the engine returns the top k candidate moves with winrate evaluations.

2. **Analysis Layer**: The models of Sections 3– 7 compute per-move and per-game metrics.

3. **Classification Layer**: The Bayesian posterior, LRT, temperature score, and profile analysis are aggregated into the final decision.

9.2 Analysis Pipeline

For each move: (1) send position to engine; (2) receive top k evaluations; (3) compute Δ, C, and A; (4) perform tempered Bayesian update; (5) compute information-theoretic metrics; (6) accumulate temporal statistics.

Profile analysis runs asynchronously, aggregating per-game statistics across the player's history and updating the baseline CI (Equation (44)).

10 Classification Decision Framework

10.1 Multi-Signal Aggregation

Classification integrates three primary signals:

1. Bayesian posterior $P(H_1 \mid D)$ under stratified H_0.
2. Log-likelihood ratio $\log \Lambda$ from the formal LRT.
3. Boltzmann temperature score $S_\tau = 1/\hat{\tau}$.

The primary rule is:

$$\text{Classification} = \begin{cases} \textbf{Cheater} & P(H_1 \mid D) \geq \theta_{\text{cheat}}, \\ \textbf{Suspicious} & \theta_{\text{susp}} \leq P(H_1 \mid D) < \theta_{\text{cheat}}, \\ \textbf{Human} & P(H_1 \mid D) < \theta_{\text{susp}}, \end{cases} \tag{51}$$

with defaults $\theta_{\text{cheat}} = 0.70$ and $\theta_{\text{susp}} = 0.30$. Agreement between the LRT rejection and the Bayesian conclusion strengthens the classification; disagreement warrants human review.

10.2 Profile-Enhanced Classification

When a profile baseline is available, the single-game posterior is supplemented by the profile test statistic Z (Equation (47)). The combined evidence is evaluated as follows:

- If the single-game posterior is borderline (Suspicious) but $Z > z_{0.05} = 1.645$, the classification is escalated to Suspicious–Elevated.
- If the single-game posterior is below θ_{susp} but $Z > z_{0.01} = 2.326$, the game is flagged for manual review.
- A confirmed Human classification requires both sub-threshold posterior and $Z < z_{0.05}$.

Single-game analysis without a profile baseline should be treated as *preliminary screening* only, with formal conclusions deferred until sufficient game history is available (Table 5).

11 Preliminary Experimental Observations

We report preliminary experiments conducted to validate design choices and identify failure modes. These experiments are exploratory in nature and do not constitute a full empirical evaluation; systematic benchmarking with labeled datasets is planned as future work.

11.1 Experimental Setup

A prototype implementation was developed integrating the Rapfi engine (via the Gomocup protocol) as the evaluation oracle. Analysis was conducted on individual games using the V2 anti-cheat analysis pipeline. Model comparison (Exponential, Gamma, Weibull) and EM mixture fitting were enabled.

11.2 Case Study: Professional Player (False Positive Scenario)

Setup. We analyzed two successive runs on the same game played by a professional human player (confirmed ground truth: no engine assistance, 22 moves analyzed, $\lambda_{\text{human}} = 10$ in Run 1, $\lambda_{\text{human}} = 50$ in Run 2).

Table 6: Summary statistics for the professional player case study.

Metric	Run 1 ($\lambda_{\text{human}} = 10$)	Run 2 ($\lambda_{\text{human}} = 50$)
Analyzed moves	22	22
Mean Δ	0.8%	0.8%
$\hat{\lambda}_{\text{MLE}}$	113.32	113.32
Near-optimal %	90.9%	90.9%
$P(\text{Cheat})$	50.3%	45.2%
Classification	Suspicious	**Human**
Confidence	73.3%	73.3%
$\log \Lambda$	22.46	22.46
$\hat{\tau}$	0.0454	0.0454
S_τ	22.02	22.02

Results.

Observations.

1. **False positive at $\lambda_{\text{human}} = 10$:** Run 1 misclassified the professional player as Suspicious. The posterior shifted correctly to Human in Run 2 after calibrating λ_{human} to 50, confirming that the null hypothesis parameter must reflect the player's actual skill tier.

2. **Inconsistency between $\log \Lambda$ and posterior:** The log-likelihood ratio $\log \Lambda = 22.46$ is extremely large (corresponding to $\Lambda = e^{22.46} \approx 5.7 \times 10^9$), yet the posterior $P(\text{Cheat}) \approx$ 45%–50% remains near the decision boundary. Sensitivity analysis shows that for all prior values tested, the posterior exceeds 99%. This suggests an implementation inconsistency between the log-LR computation and the Bayesian update, likely caused by the tempered likelihood suppressing evidence via positional complexity. This warrants investigation and is consistent with the observation that complex positions (many with $C \ll 1$) may over-discount genuine evidence.

3. **Weibull consistently wins model selection:** Across all runs, the Weibull distribution achieves the lowest AIC and BIC with shape parameter $k \approx 0.36$, indicating heavy-tailed behavior (rare but large errors). However, all three models are rejected by the KS test ($p < 0.05$), which is expected at $n = 22$ due to low test power. Reliable model validation requires $n \geq 50$.

4. **EM-fitted λ_g is engine-level:** The EM algorithm consistently fits a "good play" component with $\lambda_g \approx 6{,}800$–$7{,}600$ (corresponding to $\mathbb{E}[\Delta_g] \approx 0.013\%$–$0.015\%$), far exceeding the cheater default $\lambda_{\text{cheater}} = 50$. This reflects the professional player's near-perfect execution in the majority of moves and underscores the overlap between the high end of the human distribution and engine-assisted play.

5. **Sample size insufficiency:** The confidence score of 73.3% indicates borderline data sufficiency ($n = 22 < n_0 = 30$). Extrapolating from Section 8.4, reliable proportion-based inference requires $n \geq 44$ moves per game. Single-game analysis at $n \leq 30$ should be treated as preliminary.

Implication for Framework Design. This case study directly motivates three design revisions incorporated into the present version of the framework: (i) stratified null hypothesis (Section 3.3), (ii) profile-based baseline estimation (Section 8), and (iii) increased data-sufficiency threshold n_0 in the confidence formula (Section 5).

11.3 Case Study: Opponent-Difficulty Confound

Setup. We analyzed a complete game (21 moves of Black analyzed, confirmed ground truth: no engine assistance) to investigate the effect of opponent weakness on the complexity score. The opponent was a significantly weaker player who lost decisively by move 33.

Table 7: Summary statistics for the opponent-difficulty case study.

Metric	Value
Analyzed moves	21
Mean Δ	3.1%
$\hat{\lambda}_{\text{MLE}}$	31.31
Near-optimal %	66.7%
$P(\text{Cheat})$	0.0%
Classification	**Human**
Confidence	70.0%
$\log \Lambda$	-4.48
$\hat{\tau}$	0.1262
S_τ	7.92

Results.

Observations.

1. **Correct classification despite opponent weakness:** The framework correctly classified this game as Human ($P(\text{Cheat}) = 0.0\%$), driven primarily by two large-delta moves: Move #23 ($\Delta = 19.7\%$) and Move #15 ($\Delta = 17.5\%$). These blunders are characteristic of human play and strongly favor H_0.

2. **Opponent-difficulty confound identified:** Inspection of the move-level log reveals a qualitative pattern directly motivating Section 4.4. From Move #13 onward, the opponent's best available winrate dropped below 0.3% (Quality: Winning), yet multiple positions in the range Move #13–#33 were assigned raw complexity $C \approx 0.47$–0.49 based on candidate variance alone. Without C_{adj}, the nine near-optimal moves in the winning phase would contribute substantial spurious evidence toward H_1 in games with lower Δ overall.

3. **Trivial Position Override activates late:** The Override (Proposition 1) activated only at Move #35 onward, once w^* exceeded 0.999. Between Move #13 and Move #35, a gap of eleven analyzed moves, the opponent had effectively lost but the Override was not triggered. This confirms the need for the Opponent Difficulty Factor as a complementary mechanism that operates on *opponent* winrate rather than *player* winrate.

4. **Log LR is negative:** $\log \Lambda = -4.48$ indicates that the data are $e^{4.48} \approx 88$ times *more likely* under H_0 than H_1, consistent with correct Human classification. This contrasts sharply with the professional player case study ($\log \Lambda = 22.46$), demonstrating that the presence of genuine blunders ($\Delta \geq 15\%$) is a powerful discriminating signal.

5. **EM identifies human-like blunder component:** The EM algorithm fitted $\lambda_g = 9334$ (near-perfect moves) and $\lambda_b = 10.69$ ($\mathbb{E}[\Delta_b] = 9.4\%$, genuine blunders), with mixing weight $\pi = 0.666$. The lower π relative to the professional player case ($\pi = 0.862$) quantifies the higher blunder rate and is consistent with the ground-truth classification.

Implication for Framework Design. This case study confirms that the Opponent Difficulty Factor (Section 4.4) is necessary as a complement to the Trivial Position Override. The Override handles the extreme case of $w^* > 0.95$ from the player's perspective; the Opponent Difficulty Factor handles the intermediate case where the opponent is functionally defeated but the player's winrate has not yet reached the Override threshold. Together they provide comprehensive suppression of opponent-induced false-positive inflation.

12 Discussion

12.1 Strengths

1. **Mathematical rigor.** Every component has a clear probabilistic interpretation grounded in established statistical theory.
2. **Complexity awareness.** The framework reduces false positives from forced sequences and false negatives from selective cheating.
3. **Skill stratification.** The stratified null hypothesis addresses the fundamental skill-confounding problem, enabling application across the full spectrum of player abilities.
4. **Profile analysis.** Multi-game aggregation provides statistically sound, sample-size-justified conclusions that single-game analysis cannot support.
5. **Opponent-difficulty correction.** The Opponent Difficulty Factor prevents false-positive inflation when the analyzed player faces a weak opponent, addressing a source of bias absent from prior frameworks.
6. **Modularity and transparency.** Each component can be independently validated, updated, or replaced.

12.2 Limitations

1. **Engine dependency.** Analysis quality is bounded by the engine's evaluation accuracy. Positions where the engine errs produce misleading delta values.
2. **Parameter sensitivity.** Default parameters are heuristically chosen. EM calibration and sensitivity analysis partially address this, but full calibration requires large labeled datasets.
3. **Independence assumption.** Conditional independence of deltas may be violated when strategic plans span multiple moves. An HMM extension (Section 13) relaxes this.
4. **Selective cheating.** Sparse engine use (e.g., only at critical moves) may evade detection thresholds; the temporal model partially addresses this.
5. **Information overlap.** KL divergence and winrate delta are not fully independent (Remark 7); over-reliance on correlated metrics may produce overconfident conclusions.
6. **Stationarity.** Profile analysis assumes approximately stationary skill levels (Remark 8). Structural breaks from genuine improvement may mimic the onset of cheating.
7. **Opponent difficulty calibration.** The reference threshold $w_{\text{ref}} = 0.40$ and the minimum floor $\alpha_{\min} = 0.10$ in the Opponent Difficulty Factor (Equation (25)) are heuristically chosen. Empirical calibration from labeled game data is required; in particular, w_{ref} may need to be game-phase dependent (e.g., lower in the opening than in the endgame).

8. **Tier boundary ambiguity.** The overlap between T_4 (professional) and H_1 (engine-assisted) distributions represents a fundamental statistical limit: it may not be possible to reliably distinguish top professional play from engine assistance in a small number of games.

12.3 Ethical Considerations

Anti-cheating systems carry inherent risks of false accusations. We advocate:

1. Statistical evidence should inform human reviewers, not replace human judgment.
2. Formal action should require high posterior probability, high confidence (sufficient data), and profile-level corroboration.
3. Players should have access to the evidence and methodology used.
4. The system should be audited regularly for skill-level and demographic biases.

13 Conclusion and Future Work

We have presented a comprehensive Bayesian statistical framework for detecting engine-assisted play in online Gomoku competitions. The framework integrates parametric move-quality modeling, context-aware complexity weighting, tempered Bayesian updates, information-theoretic divergence measures, temporal switch-point detection, player skill stratification, and multi-game profile analysis into a coherent probabilistic system.

Preliminary experiments reveal a fundamental *skill-confounding problem*: without stratification, professional players are incorrectly flagged at rates comparable to engine-assisted players. The profile-based framework developed in Section 8 provides a statistically principled resolution, with sample size requirements derived from confidence-interval theory. Additionally, we identify and formalize the *opponent-difficulty confound*: positions made trivially easy by a weak opponent generate spurious evidence toward H_1 under one-sided analysis. The Opponent Difficulty Factor (Section 4.4) corrects for this bias by weighting evidential contribution by the opponent's competitive viability.

13.1 Future Work

- **Hidden Markov Model for selective cheating:** A two-state HMM with states $\{\text{Honest}, \text{Cheating}\}$ and emission distributions F_{z_t}:

$$z_t \mid z_{t-1} \sim \text{Categorical}(A_{z_{t-1}}), \quad \Delta_t \mid z_t \sim F_{z_t}, \tag{52}$$

  where A is the transition matrix. This subsumes switch-point detection as a special case [Rabiner, 1989].

- **Empirical calibration:** Fitting all model parameters via EM, MCMC, or variational inference on large labeled datasets. KS and Anderson–Darling tests should validate distributional assumptions. This includes calibration of w_{ref} and $\alpha_{\min}$ in the Opponent Difficulty Factor across different game phases and competitive contexts.

- **Systematic empirical evaluation:** Benchmarking on three scenario types—confirmed human games, engine self-play, and suspected games—with full ROC curve, AUC, precision–recall, and confusion matrix reporting.

- **Tier boundary calibration:** Determining empirically where the T_4/H_1 boundary lies using paired professional-play and engine-play datasets.

- **Dirichlet Process deployment:** Replacing the fixed two-component mixture with the DPMM (Section 3.7) in production.
- **Adversarial robustness:** Analyzing resilience against adaptive cheaters who intentionally introduce errors.
- **Generalization:** Adapting the framework to Go, Othello, Connect6, and other abstract strategy games.
- **Timing integration:** Incorporating move timing as an additional signal via a multimodal Bayesian network.

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